

# Psychophysiology, eye-tracking and VR: exemplary study design

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## ABSTRACT

In our poster we present the design of the study, which will be conducted in the Virtual Environment with the usage of psychophysiology (EDA, HR and pupil size) and eye-tracking. Although many papers implementing aforementioned methods have been carried out to date, we believe that these techniques are still not fully utilized. Therefore, the focus in our work is on the methodological steps we have taken and technical topography we created to provide proper ground for the measurement. Our aim was to increase the signal-to-noise ratio, achieve higher confounders control and obtain reliable results.

**Index Terms:** Virtual Reality (VR)—bot—avatar—eye-tracking psychophysiology;

## 1 INTRODUCTION

Since the introduction of Virtual Reality (VR) as the tool for psychological experiments [7] researchers have been working on the development of reliable measures of experience, both declarative (questionnaires) and indirect ones, including physiological (psychophysiology) and attention-based (eye-tracking). The discussion on the usability of psychophysiology [1] and eye-tracking [4] measurement in VR is still ongoing. Although implementation of these measures may increase quality and reliability of the findings [4, 14], to use them efficiently the experiment needs to meet several preconditions such as proper stimulus presentation time and dependent measure selection [3].

This work is an example of psychophysiology and eye-tracking measurement adaptation to VR psychological experiment. We describe the methodological steps taken to implement two complex measurements into the virtual character assessment and demonstrate the technical architecture of the experimental setup.

The procedure described in this poster is a part of the larger experiment which aims to investigate the relationship between realism of virtual character, [9] copresence [10], psychophysiology indicators (EDA, HR and pupil size) as well as attention allocation. During the study, participants will wear HTC Vive Pro goggles and watch the virtual characters while physiological and eye-tracking measurements are conducted. The experimental stimuli consist of 18 combinations of 3 levels of character's photographic realism, 3 levels of character's behavioral realism and 2 levels of character's interaction realism, each imposed on 10 virtual character models.

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As a result we will use 180 different stimuli to be presented to the participants.

## 2 METHODOLOGICAL STEPS

In the following section we discuss some of the methodological steps taken to create proper ground for physiological and eye-tracking measurement.

**Phasic measurement** - we decided to use a phasic component of physiological measurement, being a fast, event-related part of the signal which allows for finding correspondence between the stimulus onset and the bodily response [3]. Owing to its rapidness, the usage of phasic response enables several repetitions of each stimuli, which would be impossible with tonic measurement due to time constraints of VR usage [5].

**Repetitions of the stimuli** - inherently, the psychophysiological signal has a random error involved. Based on the central limit theorem, if we repeat the presentation of each stimulus and average the measurements over trials we get closer to the true value of the signal [3]. Therefore, to increase signal-to-noise ratio and achieve higher confounders control we iterated each of the 18 realism combinations 10 times.

**Stimulus presentation time adjustment** - to detect the phasic reaction to the stimuli we decided to display each of the virtual characters for 6 seconds following methodology used by Bradley and associates (2008). The presentation time is based on the temporal aspects of the phasic HR, EDA and pupillometry [2, 3].

**Variation in an interstimulus interval** - in the experiments involving repetitive presentation of the stimuli in tight temporal order, the physiological anticipatory reaction (physiological response while awaiting the stimuli) is expected [13]. To limit its effect on the results, we decided to vary the interstimulus interval in ranges between 9 and 12 seconds (at random) - so as to start the stimulus presentation in different moments of the anticipatory reaction and then average the physiological response over trials. The width of the ranges was based on the typical time needed for the signal to go back to the baseline levels [2]. This solution prevents overlap of the response with preceding and subsequent stimuli [8].

**Area-of-Interest (AOI) analysis in eye-tracking** - AOI are regions of the stimulus that the researcher is interested in (e.g. particular part of a landscape) and can be used to quantify how much participants gazed at the particular area [6]. We decided to make use of AOI, because it allows for precise transformation of the fixations into numeric representations which can be statistically compared between the particular regions [6]. To make AOI demarcation easier we limited the virtual character's movement and created the approximate bounding-boxes [12] for the head and rest of the body (fig. 1 in sup. mat.).

**Variation in the stimuli features** - when stimulus is presented repetitively, as in our procedure, the habituation process occurs [11]. To decrease that, we created 10 models differing in terms of secondary features such as: facial features and clothing, on which 18 different combinations of the realism were imposed (fig. 2 in sup.

mat.).

### 3 THE COMPONENTS OF THE SYSTEM

**Application for presentation of bots.** The application for obtaining feedback on the presented bots was implemented using Unity 2020.2.1f1. Virtual characters will be presented to the user in random order, alternately with fixation screen (white cross-shaped object on the black background). Exposition of each variant of bot lasts 6 seconds, while the duration of fixation point is drawn from the range 9 to 12 seconds. For the experiment the ability to freely rotate view with HMD-mounted head was turned off by appropriate Unity XR project setting.

**VR headset.** In order to provide high quality immersion to the user the visual part is served via HTC Vive Pro.

**External sensors.** To measure the pupil size and attention allocation we used Pupil Labs eye-tracker which is an add-on for HTC Vive Pro headset. For physiological measurement we utilized biosignalsplux device (PLUX Wireless Biosignals, Lisbon, Portugal) with EDA and ECG sensors.

### 4 THE ARCHITECTURE OF THE SOLUTION

The overall data flow between components was visualized in fig. 3 (sup. mat.).

EDA and ECG data is obtained from biosignals hub via Bluetooth by Unity, then optionally saved to .csv or sent over researcher-specified IP and port to the LabVIEW, which visualizes the data and saves it to the .tdms file. The signals from biosignalsplux hub are acquired via Bluetooth connection with dedicated USB dongle using Unity biosignalsplux API and custom code adjusting the biosignalsplux SDK to the needs of the project with effective sampling rate efficiency at around 1kHz with Analog to Digital Converter (ADC) 16 bits resolution.

Each sample is obtained as a row of channel-corresponding integer readouts within jagged array. The ADC package of data gets extracted, decomposed and placed into the preformatted string of data. The strings are in form of separate channel-dedicated integer columns containing: sequence numbers, universal timestamps, relative timestamps of the time passed since the start of the Unity session runtime, channel-corresponding sensor signal readouts and trigger tags indicating the event/situational context of the collected biomedical data. It is ready to be optionally either continuously saved as a sample record into the session .csv file, or sent via researcher-specified IP address and port, or both saved to .csv and sent over TCP. Sending gets performed by Unity as a server to the receiving TCP Open Connection Function of the LabVIEW .vi project running as a client-side application. The LABVIEW application was created for the purposes of realtime signal data visualization. The result is .tdms file with physiological outcome data. In the future, another data processing software like MATLAB or Python can be concurrently involved to the processing pipeline.

The data from the HMD Pupil Labs Binocular Add-on 200Hz eyetracker gets acquired by Unity via USB with the Pupil Capture in the background, then the saved session raw data can be preprocessed and exported into .csv files with use of Pupil Player. The Pupil Labs eyetracker data get merged with trigger tags information accompanied with relative time Unity session runtime timestamps. During the session the VR user gets introduced with in-VR 3D animated bots while at the same time the researcher has direct preview of the current user's in-VR perspective and the state of the connection with the Pupil Labs eyetracker, with the biosignalsplux hub and with the LabVIEW. The current state of the biosignalsplux data acquisition is visualized in LabVIEW.

### 5 SUMMARY

Our work aimed at creating an adequate ground for psychophysiological and eye-tracking measurement in VR experiment. To do that

we needed to adjust the design and the stimuli characteristics as well as make implement a custom solution combining Unity, LabVIEW and software provided by the measurement devices producers. The solution described in this paper was created for virtual character assessment, but its usability is not limited to that area. The discussed aspects of the experiment design can be easily transferred to other areas of VR research, such as virtual environment design, interactions with bots etc.

### 6 ACKNOWLEDGEMENTS

This work was financed from the budget for science in the years 2019-2023, as a research project "The impact of virtually generated characters: human aspects of bots" (project number: DI2018 015848) under the Diamond Grant programme granted by the Ministry of Science and Higher Education of Poland. Work of M. Igras-Cybulska was supported by the National Science Centre, Poland 2020/04/X/ST6/00626.

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